Understanding Quantum Correlations: Bell Inequalities and Beyond

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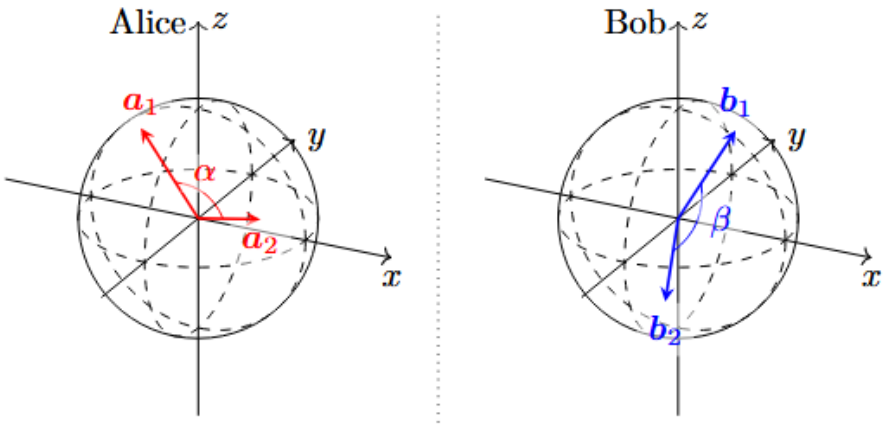
Motivation

- My initial goal was to study **open quantum systems**, which are the foundation of modern quantum technologies.
- Due to summer schedule, I worked with **Lee-san** on quantum correlations.
- Explored multiple topics: uncertainty relations, weak measurement, pseudo-joint probabilities, and **Bell inequalities**.
- Focused on Bell inequalities because they reveal fundamental **quantum nonlocality**—the phenomenon that particles can show correlations stronger than classical physics allows.
- Understanding general (n, m, d) Bell scenarios helps us:
  - Deepen knowledge of quantum correlations.
  - Support future advances in quantum computing and quantum information processing.
- **Key message:** Bell inequalities are like "rules for quantum correlations" that cannot be explained classically.

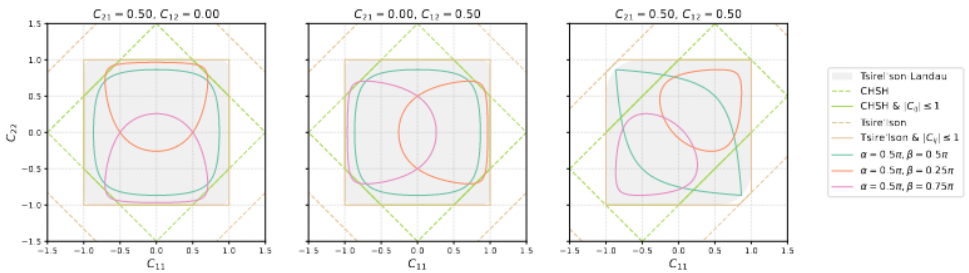
Key Idea from Lee-san's Work

- Alice and Bob each measure a quantum system with two possible settings.
- The choice of measurement directions affects the strength of correlations.
- Lee-san developed precise criteria to determine which correlations can appear in quantum mechanics.
- These criteria improve on previous classical and quantum bounds.
- **Key message:** His method allows exact characterization of allowed quantum correlations in a two-party system.

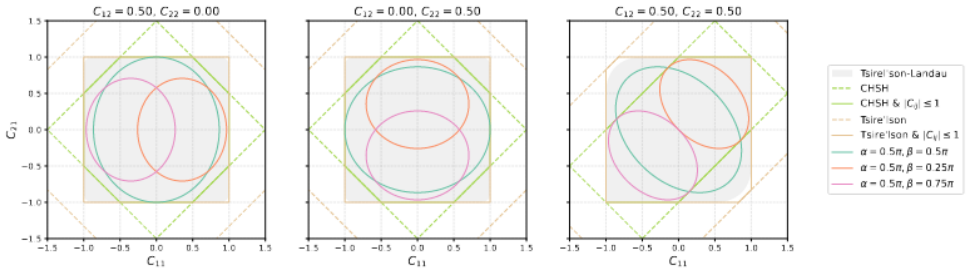


Visualizing Quantum Correlation Sets

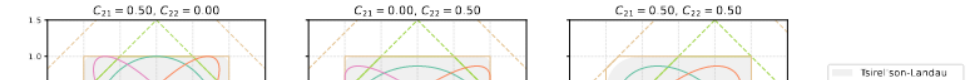
- Different curves represent allowed correlations for various measurement settings.
- Gray area: previous known quantum limits.
- Dashed boundaries: classical and Tsirel'son limits.
- **Observation:** Our inequalities exactly describe the quantum-allowed region.
- **Takeaway:** This visualization helps us intuitively see which correlations are quantum-realizable.



(a) C_{11}–C_{22} cross sections.



(b) C_{11}–C_{21} cross sections.



Why the (3,2,2) Scenario?

- It is the simplest scenario to explore **genuine three-party entanglement**.
- Allows testing of **Mermin** and **Svetlichny** inequalities, which detect three-party nonlocality.
- Helps distinguish between:
  - **Biseparable states** (only some particles are entangled)
  - **Genuine multipartite entanglement** (all three particles are entangled)
- Understanding (3,2,2) forms the basis for studying larger systems with more particles or measurement settings.

Project Workflow

- **Step 1: Generate quantum states.** Randomly choose two-qubit states and a fixed single-qubit state.
- **Step 2: Compute correlations.** Calculate three-body correlation values.
- **Step 3: Bound correlations.** Use semidefinite programming (SDP) and NPA hierarchy to determine which correlations are quantum-realizable.
- **Step 4: Visualize results.** Show distributions using scatter plots and histograms.
- **Step 5: Analyze.** Determine whether states can show genuine tripartite nonlocality and interpret physical meaning.

Biseparable States Cannot Violate Svetlichny Inequality

- Biseparable (BSP) states: only some particles are entangled.
- In (3,2,2) scenario, three-body correlations factorize into two-body + one-body contributions.
- Using known bounds from CHSH and Tsirelson inequalities:
  - Maximum Svetlichny value for BSP states cannot exceed classical limits.
- **Conclusion:** BSP states do not show genuine three-party nonlocality.

Monte Carlo Sampling of BSP States

- Randomly generate many BSP states and compute correlation values (M, S).
- Visualize the distribution:
  - Product BSP: X-shaped boundary
  - Mixed BSP: fills convex hull
- Observation: Extreme points rarely occur; no genuine three-body nonlocality is found.

Quantum Realizability of BSP States

- Mixed BSP states are combinations of product BSP states.
- All correlation points lie within the convex hull formed by product states.
- **Implication:** The set of product BSP states fully describes allowed quantum correlations.
- **Key message:** BSP correlations are an extension of the CHSH scenario; genuine three-party nonlocality is absent.

Next Steps in Current Topic

- Extend quantum-realizability criteria to larger systems with more parties or measurement settings.
- Develop more efficient SDP/NPA methods to handle high-dimensional correlations.
- Explore device-independent applications:
  - Randomness generation
  - Entanglement certification
- Connect Bell nonlocality with multipartite entanglement measures.

Long-Term Research Directions

- Study open quantum many-body systems and non-equilibrium properties using Bell inequalities and entanglement criteria.
- Use neural network approaches to approximate complex quantum correlations in large systems.
- Investigate how quantum nonlocality can inform error mitigation and practical quantum computing strategies.

Thank you!